Linux Compression Tools Notes

Linux provides multiple tools for compressing and decompressing files. Below are the most common ones with commands and examples.

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1. gzip / gunzip

Compress a file

```
gzip file.txt
```

👉Creates `file.txt.gz` and removes the original `file.txt`.

Decompress a file

```
gunzip file.txt.gz
```

👉Restores the original `file.txt`.

Keep original file while compressing

```
gzip -c file.txt > file.txt.gz
```

Example:

```
echo "Hello Linux" > file.txt
gzip file.txt
ls
```

Output:

```
file.txt.gz
```

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2. bzip2 / bunzip2

Compress a file

```
bzip2 file.txt
```

👉Creates `file.txt.bz2` and removes `file.txt`.

Decompress a file

```
bunzip2 file.txt.bz2
```

👉Restores the original `file.txt`.

Keep original file while compressing

```
bzip2 -k file.txt
```

👉Keeps both `file.txt` and `file.txt.bz2`.

Example:

```
echo "Linux Compression" > file.txt
bzip2 file.txt
ls
```

Output:

```
file.txt.bz2
```

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3. zip / unzip

Compress multiple files into a .zip

```
zip archive.zip file1 file2
```

👉Creates `archive.zip` containing file1 and file2.

Decompress (extract) files

```
unzip archive.zip
```

👉Extracts all files inside the zip.

List contents of zip without extracting

```
unzip -l archive.zip
```

Example:

```
echo "Hello" > file1.txt
echo "Linux" > file2.txt
zip archive.zip file1.txt file2.txt
ls
```

Output:

```
archive.zip  file1.txt  file2.txt
```

🔹 File Extensions

- `gzip` → `.gz`
- `bzip2` → `.bz2`
- `zip` → `.zip`

🔹 Quick Summary

Tool	Compress Command	Decompress Command	Extension
gzip	`gzip file.txt`	`gunzip file.txt.gz`	`.gz`
bzip2	`bzip2 file.txt`	`bunzip2 file.txt.bz2`	`.bz2`
zip	`zip archive.zip file1 file2`	`unzip archive.zip`	`.zip`

👈Use `gzip` and `bzip2` for single-file compression, and `zip` for multiple files packaged together.